Titanic.ipynb

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RAM Disk

[1] 0s
```
# install all libraries
```

[2] 1s
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

[5] 0s
```
# load the dataset -
# create df variable

# df = pd.read_csc("titanic.csv)
```

[4] 0s
```
df = sns.load_dataset("titanic")
```

[6] 0s
```
# display the first 5 rows
```

[7] 0s
```
df.head()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN

Next steps: Generate code with df | New interactive sheet

[8] 0s
```
# see all columns name
```

[9] 0s
```
df.columns
```
```
Index(['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare',
       'embarked', 'class', 'who', 'adult_male', 'deck', 'embark_town',
       'alive', 'alone'],
      dtype='object')
```

[10] 0s
```
# Undestand the dataset
```

[11] 0s
```
# rows and columns
```

[12] 0s
```
df.shape
```
```
(891, 15)
```

[13] ✓ 0s

```
# statistical summery
```

[14] ✓ 0s

```
df.describe()
```

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

[15] ✓ 0s

```
# we have 891 passengers with 15 columns / featurs
```

[16] ✓ 0s

```
# check missing values
```

[17] ✓ 0s

```
df.isnull().sum()
```

	0
survived	0
pclass	0
sex	0
age	177
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0
adult_male	0
deck	688
embark_town	2
alive	0
alone	0

dtype: int64

[18] ✓ 0s

```
# missing values in percenatage
```

[19] ✓ 0s

```
df.isnull().mean()*100
```

	0
survived	0.000000
pclass	0.000000
sex	0.000000
age	19.865320
sibsp	0.000000
parch	0.000000
fare	0.000000
embarked	0.224467
class	0.000000
who	0.000000
adult_male	0.000000
deck	77.216611
embark_town	0.224467
alive	0.000000
alone	0.000000

dtype: float64

[20] ✓ 0s

```
missing = df.isnull().sum()
```

[21] ✓ 0s

```
missing[missing > 0]
```

	0
age	177
embarked	2
deck	688
embark_town	2

dtype: int64

[22] ✓ 0s

```
# create a copy before cleaning
```

[23] ✓ 0s

```
# new variable name raw_df and clean_df
```

[25] ✓ 0s

```
raw_df = df
```

[26] ✓ 0s

```
clean_df = df
```

[27] ✓ 0s

```
# same data in df , raw_df, clean_df
```

[28] ✓ 0s

```
# Check for the duplicate records
```

[29] ✓ 0s

```
clean_df.duplicated().sum()
```

```
np.int64(107)
```

[30] ✓ 0s

```
# remove 107 duplicate records
```

[33] ✓ 0s

```
clean_df = clean_df.drop_duplicates()
```

[34] ✓ 0s

```
clean_df.shape
```

```
(784, 15)
```

[35] ✓ 0s

```
# we have unique 784 customers
```

[36] ✓ 0s

```
# missing values - remove - replace ?
```

[37] ✓ 0s

```
clean_df.isnull().sum()
```

	0
survived	0
pclass	0
sex	0
age	106
sibsp	0
parch	0
fare	0
embarked	2
class	0
who	0
adult_male	0
deck	582
embark_town	2
alive	0
alone	0

dtype: int64

[38] ✓ 0s

```
# age is numerical data having 106 missing values
# replace by ? - average (median)

# mean - mean robhust to outlier
# we use the median
```

[40] ✓ 0s

```
clean_df["age"].median()
```

```
28.25
```

[42] ✓ 0s

```
#clean_df["age"] = clean_df["age"].fillna(28.25)

clean_df["age"] = clean_df["age"].fillna(clean_df["age"].median())

#subqueery
```

```
/tmp/ipykernel_982/2875217557.py:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/inc
  clean_df["age"] = clean_df["age"].fillna(clean_df["age"].median())
```

[43] ✓ 0s

```
clean_df["age"].isnull().sum()
```

```
np.int64(0)
```

[44] ✓ 0s

```
# embarked is the categorical feature
```

[45] ✓ 0s

```
# so replace with - mode
```

[46] ✓ 0s

```
clean_df["embarked"].mode()
```

	embarked
0	S

dtype: object

[48] ✓ 0s

```
clean_df["embarked"].mode()[0]
# [0]  - for indexing
```

```
'S'
```

[49] ✓ 0s

```
clean_df["embarked"] =  clean_df["embarked"].fillna(clean_df["embarked"].mode()[0])
```

```
/tmp/ipykernel_982/2864406080.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/inc
  clean_df["embarked"] =  clean_df["embarked"].fillna(clean_df["embarked"].mode()[0])
```

[50] ✓ 0s

```
clean_df["embarked"].isnull().sum()
```

```
np.int64(0)
```

[51] ✓ 0s

```
# same for embarked town
```

[52] ✓ 0s

```
clean_df["embark_town"] =  clean_df["embark_town"].fillna(clean_df["embark_town"].mode()[0])
```

```
/tmp/ipykernel_982/1537302128.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/inc
  clean_df["embark_town"] =  clean_df["embark_town"].fillna(clean_df["embark_town"].mode()[0])
```

[53] ✓ 0s

```
clean_df["embark_town"].isnull().sum()
```

```
np.int64(0)
```

[54] ✓ 0s

```
# we drop the dec column
```

[55] ✓ 0s

```
clean_df = clean_df.drop(columns=["deck"])
```

[56] ✓ 0s

```
clean_df.isnull().sum()
```

	0
survived	0
pclass	0
sex	0
age	0
sibsp	0
parch	0
fare	0
embarked	0
class	0
who	0
adult_male	0
embark_town	0
alive	0
alone	0

dtype: int64

[57] ✓ 0s

```
# Data Types ?
```

[58] ✓ 0s

```
clean_df.dtypes
```

	0
survived	int64
pclass	int64
sex	object
age	float64
sibsp	int64
parch	int64
fare	float64
embarked	object
class	category
who	object

adult_male	bool
embark_town	object
alive	object
alone	bool

dtype: object

[59] ✓ 0s
```
# conver object data type into category
```

[60] ✓ 0s
```
clean_df["sex"] = clean_df["sex"].astype("category")
```

[61] ✓ 0s
```
clean_df["embarked"] = clean_df["embarked"].astype("category")
```

[62] ✓ 0s
```
clean_df["who"] = clean_df["who"].astype("category")
```

[65] ✓ 0s
```
clean_df["embark_town"] = clean_df["embark_town"].astype("category")
```

[66] ✓ 0s
```
clean_df["alive"] = clean_df["alive"].astype("category")
```

[67] ✓ 0s
```
clean_df.dtypes
```

	0
survived	int64
pclass	int64
sex	category
age	float64
sibsp	int64
parch	int64
fare	float64
embarked	category
class	category
who	category
adult_male	bool
embark_town	category
alive	category
alone	bool

dtype: object

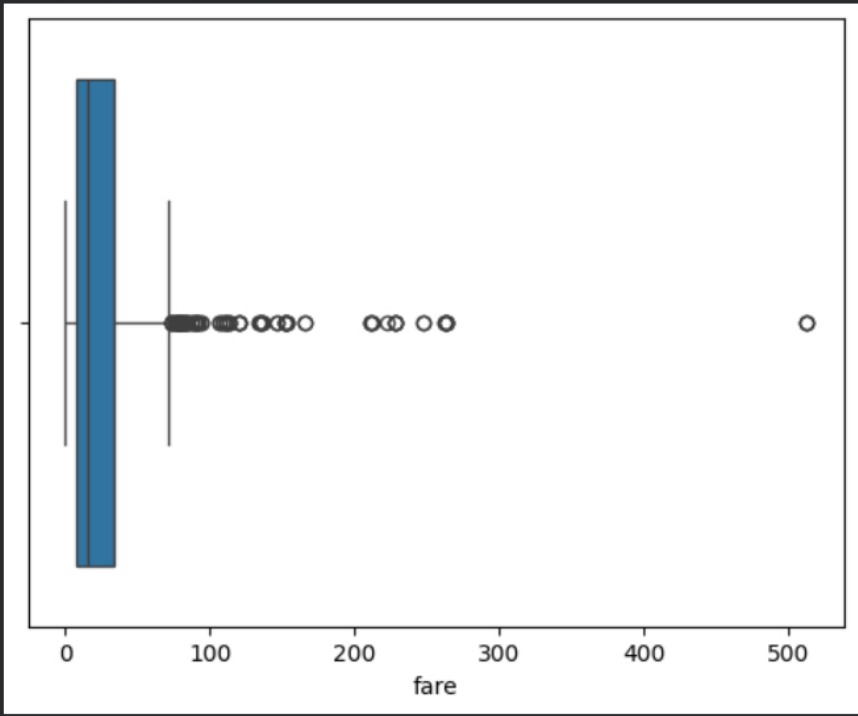
[68] ✓ 0s
```
# check for the outliers
```

[69] ✓ 0s
```
# 5 number summery - box plot -
# Q1, Q2, Q3, IQR, LB, UB
```

[70] ✓ 0s

```
sns.boxplot(data= clean_df, x= "fare")
```

```
<Axes: xlabel='fare'>
```



[71] ✓ 0s

```
Q1 = clean_df["fare"].quantile(0.25)
```

[72] ✓ 0s

```
Q2 = clean_df["fare"].quantile(0.50)
```

[73] ✓ 0s

```
Q3 = clean_df["fare"].quantile(0.75)
```

[74] ✓ 0s

```
IQR = Q3 - Q1
```

[75] ✓ 0s

```
lower_limit = Q1 - 1.5 * IQR
```

[77] ✓ 0s

```
upper_limit = Q3 + 1.5 * IQR
```

[78] ✓ 0s

```
outliers = clean_df[
    (clean_df["fare"] < lower_limit) |
    (clean_df["fare"] > upper_limit)
]
```

[79] ✓ 0s

```
len(outliers)
```

```
102
```

[82] ✓ 0s

```
# data cleaning
clean_df["sex"].unique()
```

```
['male', 'female']
Categories (2, object): ['female', 'male']
```

[83] ✓ 0s

```
clean_df["sex"] = clean_df["sex"].astype("string").str.upper()
```

[84]
```
clean_df["sex"].unique()
```

```
<StringArray>
['MALE', 'FEMALE']
Length: 2, dtype: string
```

[85]
```
# feature engg
```

[86]
```
clean_df["age_group"] = pd.cut(
    clean_df["age"],
    bins = [0,12,18,35,60,100],
    labels =["child", "teen", "adult", "senior", "oldage"]
)
```

[87]
```
clean_df["age_group"].unique()
```

```
['adult', 'senior', 'child', 'teen', 'oldage']
Categories (5, object): ['child' < 'teen' < 'adult' < 'senior' < 'oldage']
```

[89]
```
clean_df.head()
```

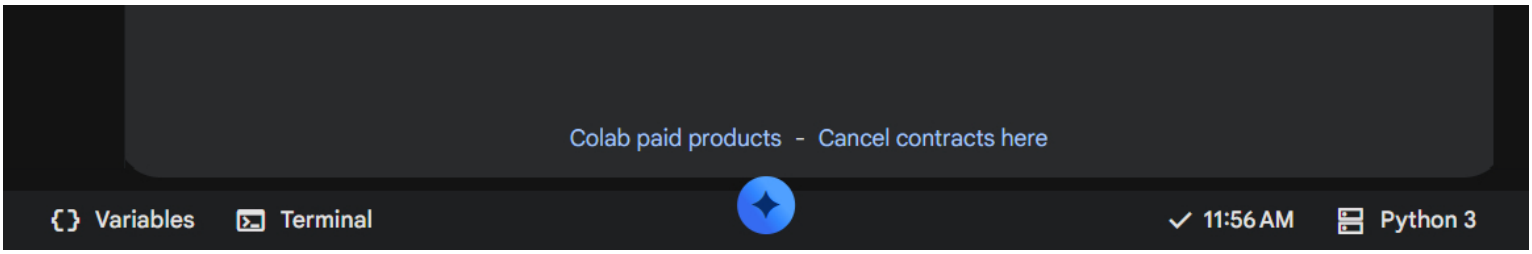
	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	emb
0	0	3	MALE	22.0	1	0	7.2500	S	Third	man	True	So
1	1	1	FEMALE	38.0	1	0	71.2833	C	First	woman	False	
2	1	3	FEMALE	26.0	0	0	7.9250	S	Third	woman	False	So
3	1	1	FEMALE	35.0	1	0	53.1000	S	First	woman	False	So
4	0	3	MALE	35.0	0	0	8.0500	S	Third	man	True	So

Next steps: Generate code with clean_df | New interactive sheet

[90]
```
clean_df["family_size"]= clean_df["sibsp"] + clean_df["parch"]+1
```

[91]
```
# EDA - exploratory data analysis
```

[ ]
```
Start coding or generate with AI.
```


Colab paid products - Cancel contracts here
Variables
Terminal
11:56 AM
Python 3